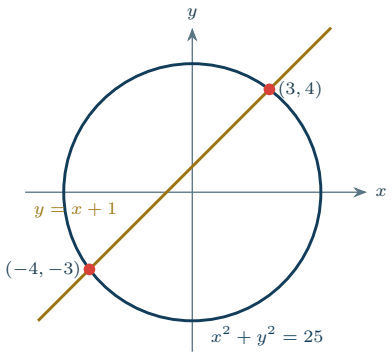
ALGEBRA ON ONE PAGE

When simultaneous equations give a quadratic

Substitute a line into a curve of degree two and the crossings become the roots of a quadratic, counted by the discriminant.

1 / THE RULE AND THE PICTURE

Substitute the linear equation into the curved one, and solve the quadratic that results



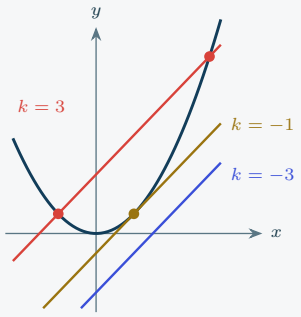
To solve $y = x + 1$ together with $x^2 + y^2 = 25$, replace y in the circle:

$$x^2 + (x+1)^2 = 25 \implies 2x^2 + 2x - 24 = 0 \implies x^2 + x - 12 = (x+4)(x-3) = 0.$$

The roots $x = 3$ and $x = -4$ give $y = 4$ and $y = -3$ from the line, and both points satisfy the circle ($9 + 16 = 25$ and $16 + 9 = 25$). The number of crossings is the number of real roots, which the discriminant $\Delta = b^2 - 4ac$ decides: here $\Delta = 1 + 48 = 49$.

2 / LINE AND PARABOLA / A FAMILY OF PARALLEL LINES

The lines $y = 2x + k$ meet $y = x^2$ twice, once or never as k falls



Setting $x^2 = 2x + k$ gives $x^2 - 2x - k = 0$ with $\Delta = 4 + 4k$. At $k = 3$ the roots are $x = -1$ and $x = 3$, so the points are $(-1, 1)$ and $(3, 9)$. At $k = -1$ the equation is $(x - 1)^2 = 0$, a double root, so the line touches the parabola at $(1, 1)$ and is the tangent. At $k = -3$ we get $\Delta = -8 < 0$ and no real crossing.

3 / SUM AND PRODUCT / A SYMMETRIC PAIR

The system $x + y = s$, $xy = p$ gives $t^2 - st + p = 0$, whose roots are x and y

Substituting $y = s - x$ into $xy = p$ gives $x(s-x) = p$, that is $x^2 - sx + p = 0$. For $x + y = 5$ and $xy = 6$ this is $x^2 - 5x + 6 = (x-2)(x-3) = 0$, so $(x, y) = (2, 3)$ or $(3, 2)$, and the symmetry of the pair means that both orders appear. When $\Delta = s^2 - 4p < 0$ the line misses the hyperbola: $x + y = 2$ and $xy = 2$ give $x^2 - 2x + 2 = 0$ with roots $1 \pm i$.

4 / WHY A QUADRATIC / COUNTING DEGREES

A linear expression inside a degree two equation leaves degree two in one unknown

curve	after substituting	Δ	crossings
circle $x^2 + y^2 = 25$, line $y = x + 1$	$x^2 + x - 12 = 0$	49	2
parabola $y = x^2$, line $y = 2x - 1$	$x^2 - 2x + 1 = 0$	0	1 (tangent)
parabola $y = x^2$, line $y = 2x - 3$	$x^2 - 2x + 3 = 0$	-8	0
hyperbola $xy = 6$, line $x + y = 5$	$x^2 - 5x + 6 = 0$	1	2

Two lines give a linear equation after substitution, hence one crossing, a line and a conic give a quadratic and so at most two, and in general a line meets a curve of degree n in at most n points. Tangency is the double root.

5 / TANGENT TO A CIRCLE / WHEN DOES THE QUADRATIC HAVE A DOUBLE ROOT

The line $y = mx + k$ touches $x^2 + y^2 = r^2$ exactly when $k^2 = r^2(1 + m^2)$

Substituting gives $(1 + m^2)x^2 + 2mkx + k^2 - r^2 = 0$ with $\Delta = 4\big(r^2(1 + m^2) - k^2\big)$, which vanishes at $k^2 = r^2(1 + m^2)$, the condition that the distance $|k|/\sqrt{1 + m^2}$ from the centre equals r. For $r = 5$ and $m = 1$ the tangents are $y = x \pm 5\sqrt{2} \approx x \pm 7.07$, so the line $k = 1$ cuts the circle, with $\Delta = 4(50 - 1) = 196$ and roots $x = \dfrac{-2 \pm 14}{4} = 3,\ -4$.

ASIDE / COMPLEX ROOTS

A negative discriminant means no real crossing; the complex roots are crossings that the real plane cannot show.

ASIDE / CHECK IN PYTHON

`np.roots([1,1,-12])` gives 3 and -4, and `np.roots([1,-2,2])` gives $1 \pm 1j$.

Convention. Substitute the line into the curve, collect to $ax^2 + bx + c = 0$, and read the crossings from $\Delta = b^2 - 4ac$.

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